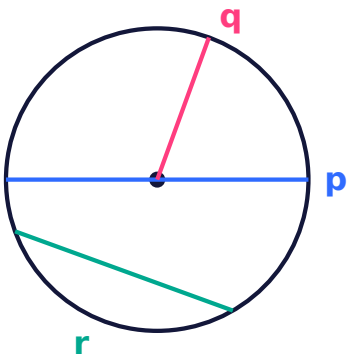
GEOMETRY QUEST · WORLD 7

Circles & π

ANSWER SHEET · for the grown-up

Answers are shown in green. “Check” means look at the child’s drawing or measuring: allow ±2° or ±2 mm.

1 Which line is the radius, which is the diameter, and which is a chord?



radius: **q**

diameter: **p**

chord: **r**

2 Fill in: a radius of 4 cm means a diameter of ___ cm. A diameter of 15 cm means a radius of ___ cm.

8 cm **7.5** cm

3 Circumference = π × diameter. Use π ≈ 3.14. Find the circumference when the diameter is 10 cm, and when it is 50 cm.

31.4 cm **157** cm

4 Area = π × radius × radius. Find the area when the radius is 5 cm, and when it is 10 cm.

78.5 cm² **314** cm²

5 A bike wheel is 70 cm across. How far does the bike travel in 10 full turns of the wheel? Give your answer in metres.

≈ 22 m (2198 cm)

6 Kitchen experiment: wrap string round a tin, measure it, then measure across the tin. Divide the first by the second.

Around: _____

Across: _____

Around ÷ across = **≈ 3.1**

Adapted from Geometry for Elementary School, Wikibooks, CC BY-SA 3.0